

Operations Research 2017/2018

Exercises Sheet nº1

LP model formulation and graphical representation

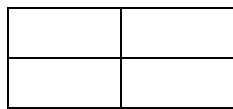
1. Formulate the LP model for each one of the following examples:
 - a) A carpenter has got 6 pieces of wood and he is available 28 hours per month to manufacture two different types of benches. Each bench of type I requires 2 pieces of wood and demands 7 working hours. Each bench of type II requires 1 piece of wood and demands 8 working hours. The unitary profit resulting from selling the benches are, respectively, 120 and 80 U.M. (Monetary Units), for type I and type II. How many wooden benches of each type must the carpenter manufacture per month, if he wants to maximize the profit? (Considers that he sells every bench that he manufactures).
 - b) A butcher shop prepares its traditionally meat balls, mixing cow meat and pork meat. The cow meat contains 20% of fat and costs 12 U.M./kg. The pork meat contains 32% of fat and costs 6 U.M./kg. Which amount of cow meat and of pork meat should contain each kilogram of meat balls, in order to minimize costs and to maintain the quantity of fat substance never superior to 25%?
 - c) In the emergence service of an hospital, that works 24 hours per day, the minimum requirements for the nursing staff in the different hours of the day are the following ones:

<u>Hours</u>		<u>Minimum requirements</u>
0 - 4	→	4
4 - 8	→	6
8 - 12	→	10
12 - 16	→	8
16 - 20	→	12
20 - 0	→	6

Each nurse works 8 consecutive hours per day and turns are initiated each 4 hours starting at 0 (thus, turns start at 0, 4, 8, 12, 16, 20 and 0 again). Formulate the problem that determines the minimum number of nurses that hospital should have, in order to satisfy the previous requirements.

(Consider that each nurse works in a single turn and that this turn is always the same.)

- d) A carpenter receives each month orders of 10000 pieces of wood with dimensions $40 \times 20 \text{ cm}^2$, 8000 pieces of wood with dimensions of $40 \times 40 \text{ cm}^2$ and 5000 pieces of wood with dimensions of $20 \times 80 \text{ cm}^2$. Given that the original pieces of wood (raw material) have dimensions of $40 \times 80 \text{ cm}^2$, there are 5 possible ways to cut the pieces in order to obtain the ordered ones, as it is shown in the figure below.



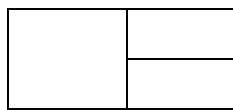
Pattern 1



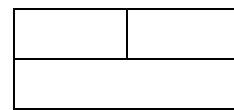
Pattern 2



Pattern 3



Pattern 4



Pattern 5

Formulate the problem that makes possible to achieve the best combination of patterns that exactly satisfy the monthly orders and minimize the amount of used raw material.

- e) A craftsman dedicates his week-ends to his favorite hobby: construction of musical instruments. Since these instruments are very difficult to find and the orders are normally made with enough antecedence, he intends to make a plan of the production of violins and violoncellos for the next year.

The wood used in the construction of these instruments must present a very good quality and specific type of wood is needed for specific parts of the musical instruments. For the arm of each instrument it is necessary to use a very rare type of wood, of which it is only possible to get 4800 cm^3 per year. Each violin needs 200 cm^3 of this wood and each violoncello needs 1600 cm^3 . It is known that, next year, the craftsman will work 48 weekends. It is also known that each violin requires 4 weekends of work and each violoncello requires 8 weekends.

The profits achieved with the construction of these instruments are of 2000 U.M. for each violin and of 8000 U.M. for each violoncello. Taking into account these conditions, how many violins and violoncellos should the craftsman produce next year in order to maximize the profit obtained from the sales. (Consider that he sells every units that he produces.)

- f) An enterprise dealing with toys produces two types of remote controlled cars (A and B). Each car of type A requires twice the time of production when compared with a car of type B. It is know that if all the cars were of type B the plant would have availability to daily

produce a maximum of 400 cars. It is also known that the average of daily sales do not exceed 150 and 200 units, respectively for cars of types A and B. Assuming that profits are of 4000 U.M and 2500 U.M., respectively for cars of types A and B, formulate the problem in order to maximize the daily profit.

- g) A factory that produces articles for hairdressers, provides 3 types of hair spray: normal fixing, strong fixing and ultra-strong fixing. For the production, the factory uses 4 base products, as indicated in the table:

Material	Amount of material (in units) necessary for the production of a unit of hair spray			Maximum amount of material (in units) available for month
	Normal	Strong	Ultra-Strong	
A	2	1	1	1500
B	1	2	3	1200
C	2	0	1	300
D	0	1	3	900

Assuming that each normal hair spray unit originates a profit of 75 U.M., each strong hair spray unit a profit of 80 U.M. and each ultra-strong hair spray unit a profit of 100 U.M., formulate the problem in order to maximize the profit obtained each month by the factory. (Consider that the factory sells all the produced units.)

- h) A rustic crockery factory has two production lines responsible for the manufacturing of two models of clay pots with a lot of demand from the market. For convenience, we will designate the two models of pots by A and B. Each time the first production line is activated one ton of raw material is consumed and 200 pots of type A, 80 pots of type B pots and 2 defective pots are produced. Each time the second production line is activated, it is consumed the same amount of raw material and 100 pots of type A, 400 pots of type B and 12 defective pots are produced. The factory has 300 tons of raw material available per month and must satisfy monthly orders of 20000 of pots type A and of 40000 pots of type B. Formulate the problem in order to minimize the amount of defective pots produced each month.
- i) A small factory of wooden toys plans to produce three new toys: trains, horses and huts. The production of these toys require carpentry and finishing specialized hand labor. The production of a train requires 1 hour of carpentry and 1 hour of finishing. The production of a horse requires 3 hours of carpentry and 2 hours of finishing. The production of a hut requires two hours of carpentry and 1 hour of finishing. The factory has 10 employees in the carpentry section and 7 in the finishing section, and each employee works 40 hours

per week. The profit per unit of trains, horses and huts is of 20€, 50€ and 25€, respectively.

The factory wants to know how many units of each type of toy should produce in order to maximize its weekly profit (assuming that the factory sells everything that it produces.)

- j) The Comerbem, a company in the food area, aims to publicize one of its new products and, thus, it hired the Bestinform firm, specialized in marketing, to organize a promotion campaign. The Comerbem wants that publicity is made via radio and via TV and has as target audience of customers from three age groups: more than 40 years, between 25 and 40 years and under 25 years of age. According to Bestinform, one minute of TV broadcast will cost 7000 € and, predictably, will reach an average of 16000 spectators in the group of above 40 years, 12500 in the group of 25-40, and 8600 in the group of sub-25. On the other hand, Bestinform estimates that one minute of advertising on Radio will reach 4500 listeners aged over 40 years, 8000 in the age group of 25-40, and 14000 in the group of under-25, and will cost 2,500 €. The Comerbem intends that the promotion campaign in its global (TV and Radio) allows a minimum daily exposure to a total of 65,000 people in the group over 40 years, a total of 80,000 people aged 25-40, and a total of 70,000 people in sub-group 25. On the other hand, it requires that all the publicity does not exceed 15 minutes per day.

To help Bestinform to perform its task, formulate the problem in terms of a linear programming model.

2. Use the **graphical method** to solve the following problems:

a) Maximize $x_1 + 2x_2$

subject to

$$\begin{aligned} x_1 &\leq 2 \\ x_2 &\leq 2 \\ x_1 - 2x_2 &\leq 3 \\ x_1 + x_2 &\leq 3 \\ x_1 \geq 0, x_2 &\geq 0 \end{aligned}$$

b) Minimize $x_1 + x_2$

subject to

$$\begin{aligned} x_1 - x_2 &\leq 2 \\ x_1 - x_2 &\geq -3 \\ x_1 \geq 0, x_2 &\geq 0 \end{aligned}$$

c) Maximize $x_1 + x_2$

subject to

$$\begin{aligned} x_1 - x_2 &\leq 2 \\ x_1 - x_2 &\geq -3 \\ x_1 \geq 0, x_2 &\geq 0 \end{aligned}$$

d) Minimize $15x_1 + 20x_2$

subject to

$$\begin{aligned} x_1 + 2x_2 &\geq 10 \\ 2x_1 - 3x_2 &\leq 6 \\ x_1 + x_2 &\geq 6 \\ x_1 \geq 0, x_2 &\geq 0 \end{aligned}$$

e) Maximize $2x_1 + 3x_2$

subject to

$$\begin{aligned} x_1 + x_2 &\leq 7 \\ 2x_1 + 3x_2 &\leq 12 \\ x_1 &\leq 5 \\ x_1 \geq 0, x_2 &\geq 0 \end{aligned}$$

f) Minimize $3x_1 + 2x_2$

subject to

$$\begin{aligned} 2x_1 + 2x_2 &\leq 8 \\ x_1 + 5x_2 &\geq 10 \\ -x_1 + 3x_2 &= 6 \\ x_1 \geq 0, x_2 &\geq 0 \end{aligned}$$

g) Maximize $3x_1 - x_2$

subject to

$$2x_1 + x_2 \geq 2$$

$$x_1 + 3x_2 \geq 3$$

$$x_2 \leq 4$$

$$x_1 \geq 0, x_2 \geq 0$$

h) Minimize $10x_1 + 20x_2$

subject to

$$-x_1 + 2x_2 \leq 15$$

$$x_1 + x_2 \leq 12$$

$$5x_1 + 3x_2 \leq 45$$

$$x_1 \geq 0, x_2 \geq 0$$